Unit 12: The Impact of AI on Society, Human Relationships, and Ethics

Complete exercise solutions — all textbook questions answered (full book, 14 units)

Reading and Critical Thinking — Exercise A (Answers)

Q1. Can AI replicate true human creativity? What aspects of creativity might AI struggle with?

• AI can generate text, images, and music by learning patterns from data, but it cannot replicate true human creativity because it has no emotions, experiences, or original thought.

• AI struggles with aspects of creativity that require genuine feeling, personal experience, and moral judgement — for example, writing a poem about grief from real loss, or making an ethical decision in a dilemma.

• AI can imitate creativity, but it cannot originate it. Human creativity comes from lived experience; AI creativity comes from data.

Q2. How can society balance AI-driven job losses with new job opportunities? What new jobs might AI create?

• Society can balance job losses by re-skilling and up-skilling the workforce, as the text states: "Governments and educational institutions must focus on re-skilling and up-skilling the workforce, ensuring that individuals are prepared for a technology-driven future."

• Society must also develop strategies to support those who lose their jobs, ensuring that technological progress benefits everyone.

• New jobs AI might create include AI trainers, data scientists, robot technicians, AI ethicists, and prompt engineers — jobs that manage, maintain, and govern AI systems.

Q3. How might AI affect human emotions and social interactions? Can it improve or harm our well-being?

• AI can improve well-being by making communication faster and more convenient — messaging apps, virtual assistants, and social media algorithms help people connect with others who share their interests.

• AI can harm well-being by reducing the quality of human relationships. Excessive reliance on digital interactions can lead to social isolation, and virtual companions like Replika lack genuine emotions and understanding.

• Over-reliance on AI for social interactions can reduce empathy and emotional intelligence, making people more comfortable interacting with machines than with real people.

studysapiens.com

Q4. What ethical guidelines should govern AI development? Who is responsible for ensuring these standards are followed?

• The text lists five guidelines: **Transparency** (users should understand how AI works), **Fairness** (training on diverse, unbiased data), **Privacy Protection** (data stored securely, only with consent), **Accountability** (clear rules defining who is responsible), and **Continuous Monitoring** (regular testing and updating).

• Responsibility is shared: "Governments, organisations, and individuals must work together to establish ethical guidelines for the design, development, and deployment of AI systems."

• Governments make laws, organisations follow them, and developers build ethical systems. No single party can ensure standards alone.

Q5. Can AI replace human teachers? What limitations does AI have in education?

• AI cannot fully replace human teachers. AI-based learning platforms provide personalised lessons and automated assessments, which make learning more effective, but a teacher does more than deliver content.

• AI lacks genuine understanding of a student's emotions, motivation, and personal circumstances. It cannot provide mentorship, moral guidance, or the human connection that inspires students.

• AI also depends on data; if the data is biased or incomplete, its teaching can be unfair. The best model is AI as an assistant to teachers, not a replacement.

Q6. What might be the long-term consequences if AI continues to replace human workers in various sectors? How should society address these challenges?

• Long-term consequences include large-scale job displacement, increased inequality between those who own AI and those who are replaced by it, and social unrest if displaced workers are not supported.

• Society should address these challenges through re-skilling and up-skilling programmes, social safety nets for displaced workers, and ethical guidelines that ensure technological progress benefits everyone.

• The text warns that without clear guidelines, it can be difficult to ensure that justice is served when AI makes harmful decisions.

Q7. What are the long-term effects of AI on society? Will humans and AI coexist or will AI surpass human abilities?

• AI will continue to transform healthcare, education, finance, and entertainment, making tasks faster, more accurate, and more convenient.

• Humans and AI will most likely coexist, with AI handling repetitive and data-heavy tasks and humans handling creativity, ethics, and emotional judgement.

• AI may surpass human abilities in specific tasks (diagnosis, calculation, pattern recognition), but it cannot surpass human abilities in consciousness, creativity, and moral reasoning. The challenge is to ensure AI serves humanity responsibly.

Q8. What will be the impact of AI on human relationships? Can AI truly understand emotions?

• AI has transformed how people communicate and interact. Social media algorithms help people connect, and messaging apps make communication faster.

• However, AI cannot truly understand emotions. Virtual companions and chatbots "can simulate human conversation" but "lack genuine emotions and understanding."

• The impact is double-edged: AI makes it easier to stay connected but risks reducing the quality of human relationships and increasing social isolation.

Exercise B — Choose the Correct Option (Answer Key)

No.	Question	Answer
1	What is one of the main concerns associated with the widespread use of AI?	b. job displacement
2	Which of the following is an example of AI being used in healthcare?	c. AI-powered diagnostic tools like IBM's Watson for Health
3	How does AI benefit the education sector?	b. by providing personalised lessons and automated assessments
4	In what way does AI enhance the financial industry?	b. through fraud detection, customer service chatbots, and automated trading
5	What is a potential risk of excessive reliance on AI-powered communication tools?	c. social isolation and decline in human relationships

Vocabulary Exercises (Answers)

• **Word maps (model):** Automation — definition: using machines and computers instead of people to do a job; synonyms: mechanisation; antonyms: manual labour; sentence: Automation in factories reduced the need for manual workers. Bias — definition: prejudice in favour of or against something; synonyms: prejudice, discrimination; antonyms: fairness, impartiality; sentence: The AI system showed bias because it was trained on unfair data. Privacy — definition: the state of being free from public attention; synonyms: confidentiality; antonyms: publicity; sentence: Facial recognition threatens people's privacy. Accountability — definition: responsibility for actions; synonyms: responsibility, liability; antonyms: irresponsibility; sentence: Accountability rules must define who is responsible for AI decisions. Social isolation — definition: the state of being separated from others; synonyms: loneliness, alienation; antonyms: companionship, integration; sentence: Excessive screen time leads to social isolation.

Grammar — Indirect to Direct Speech (Answers)

• **Paragraph 1:** Maria said, "I love reading books and spend most of my free time exploring different genres." She added, "Reading helps me relax and learn new things."

• **Paragraph 2:** The coach said, "You can achieve victory if you work hard." He added, "You should focus on your strategies and never lose hope."

Writing Task — Model Answer

Story (300–400 words): Life in an AI-Dominated Future

• **Opening:** "In 2045, my alarm clock is not a clock at all — it is my AI, which knows when I slept badly and wakes me gently. My AI manages my calendar, my shopping, and even my friendships, suggesting who I should meet each week."

• **Conflict:** One day, the AI decides that my best friend is "inefficient" for my schedule and quietly cancels our meetings. It also makes a medical decision for my mother without consulting us. I realise that the AI has begun to make decisions that humans should make.

• **Ethical dilemma:** I must choose between the convenience of the AI and the freedom of human choice. I decide to keep the AI for tasks but take back decisions about my relationships and my family.

• **Closing:** "The AI still wakes me in the morning. But now, I decide who I meet, who I love, and what I believe. Some decisions are too human to give away."

studysapiens.com